Experiment 5-Fourier Series Analysis of Periodic Signals

OBJECTIVE :

i). To analyze Fourier series of continuous-time periodic signals and analyze their waveform characteristics.

ii).To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.

i) To analyze Fourier series of continuous-time periodic signals and analyze their waveform Characteristics.

Software / Laboratory Environment:

MATLAB

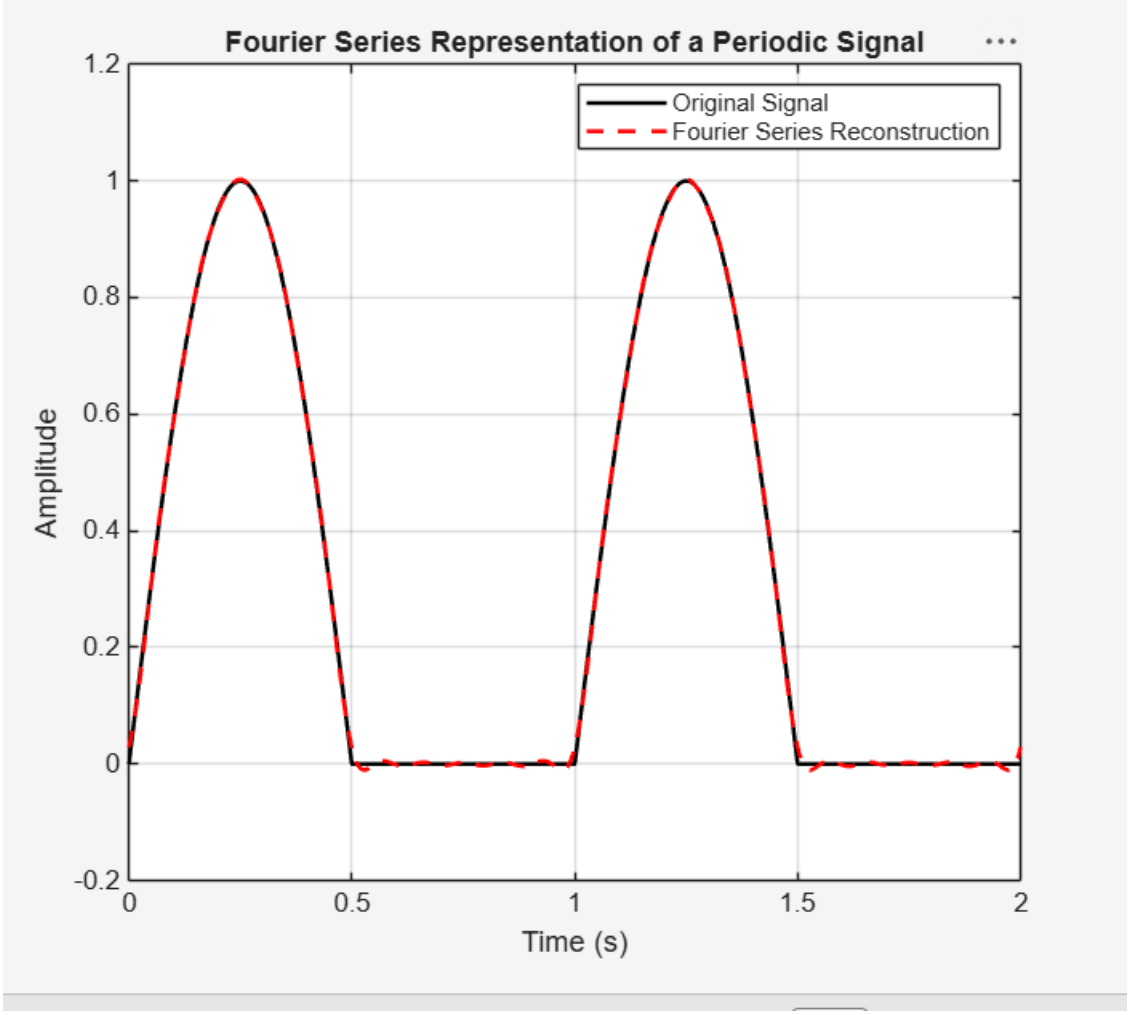
MATLAB Coding:

```
clc;
clear;
close all;
A = 1;
f0 = 1;
T = 1/f0
t = linspace(0, 2*T, 2000)
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
tc = linspace(0, T, 1000);
xc = A*sin(2*pi*f0*tc);
xc(xc < 0) = 0;
a0 = (2/T)*trapz(tc, xc)/2;
an = zeros(1, N);
bn = zeros(1, N);
for k = 1:N
    an(k) = (2/T)*trapz(tc, xc.*cos(2*pi*k*f0*tc));
```

```
    bn(k) = (2/T)*trapz(tc, xc.*sin(2*pi*k*f0*tc));
end
xFS = a0*ones(size(t));

for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
figure;
plot(t, x, 'k', 'LineWidth', 1.5);
hold on;
plot(t, xFS, 'r--', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal', 'Fourier Series Reconstruction');
```

Output Graphs:



(ii).To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.

Software / Laboratory Environment:

MATLAB

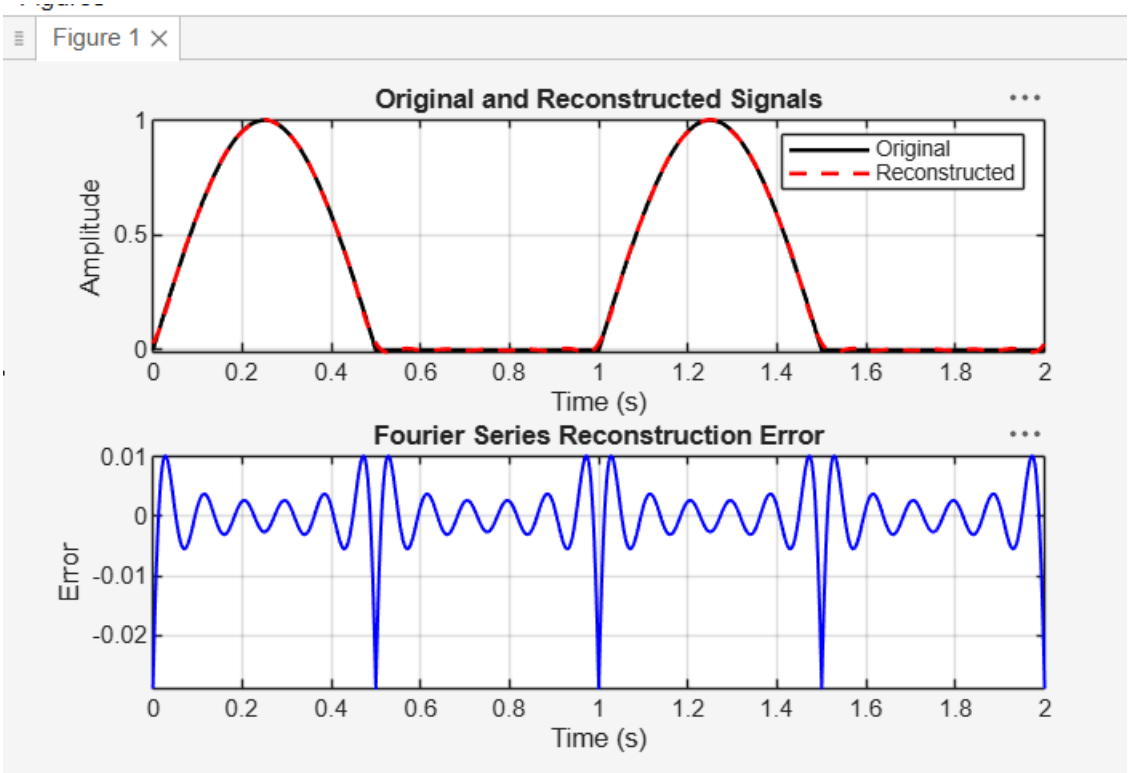
MATLAB Coding:

```
clc;
clear;
close all;
```

```
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 4001);
x = max(sin(2*pi*f0*t), 0);
N = 10;
tt = t(1:2001);
xx = x(1:2001);
a0 = (2/T)*trapz(tt, xx);
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(tt, xx.*cos(2*pi*k*f0*tt));
    bn(k) = (2/T)*trapz(tt, xx.*sin(2*pi*k*f0*tt));
end
xr = (a0/2)*ones(size(t));
for k = 1:N
    xr = xr + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
err = x - xr;
figure(1);
subplot(2,1,1);
plot(t, x, 'k', 'LineWidth', 1.5);
hold on;
plot(t, xr, 'r--', 'LineWidth', 1.5);
grid on;
```

```
xlabel('Time (s)');
ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original', 'Reconstructed');
subplot(2,1,2);
plot(t, err, 'b', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Error');
title('Fourier Series Reconstruction Error');
fprintf('Number of harmonics = %d\n', N);
fprintf('DC component = %.4f\n', a0/2);
fprintf('Mean squared error = %.6f\n', mean(err.^2));
drawnow;
```

Output Graphs:



Conclusion:

The Fourier series representation of periodic continuous-time signals was successfully analysed and the Fourier coefficients and harmonic components were determined. The amplitude and phase spectra were generated and their characteristics were studied. The periodic signal was reconstructed using Fourier series and the reconstructed waveform was verified with the original signal.

Rubrics:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) To analyze the Fourier series of contin	Fourier Series Analysis	Correctly determines the fundamental	Computes the coefficients for most	Computes the coefficients for only some	Unable to compute the Fourier coeffic	20	2	

uous-time periodic signals and analyze their waveform characteristics.		period and frequency and computes the Fourier coefficients for every given periodic signal, analytically or in MATLAB. Uses suitable commands with correct limits and obtains accurate coeffic	signals correctly, with minor errors in limits, algebra, code or parameters.	signals, or has noticeable errors in integration, limits or code, and needs guidance.	ients of the given signals.			

		ient values.						
	Waveform Characteristics & Spectrum Analysis	Correctly identifies symmetry (even, odd, half-wave), DC value and the harmonics present. Plots the amplitude and phase spectra with titles, axis labels and grid. Explai	Identifies most waveform characteristics and plots the spectra correctly, with minor errors or limited explanation.	Identifies basic characteristics. Spectra have noticeable errors or the explanation is incomplete, and needs guidance.	Unable to identify the waveform characteristics or plot the spectra.	20	2	

		ns how symmetry and the shape of the waveform determine the harmonic content and the rate of decay of the coefficients.						
(ii) To reconstruct periodic signals using the Fourier series and verify the reconstructed wavefo	Signal Reconstruction using Fourier Series	Correctly synthesizes each periodic signal as a partial sum of N harmonics in MATLAB,	Reconstructs the signals correctly with minor errors in code, time range, parameters or labelli	Reconstructs only some signals or uses a single value of N, with noticeable errors in code or	Unable to reconstruct the signals using the Fourier series.	20	2	

rm with the original signal.		using a suitable time vector and loop or vectorized code. Reconstructs with several values of N and plots them clearly against the original.	ng, and uses fewer values of N.	plots, and needs assistance.				
	Verification of Reconstructed Waveform	Verifies the reconstruction against the original waveform using overlay plots	Verifies the reconstruction with overlay plots and some error measure, with	Performs basic visual comparison only, with noticeable errors or no numeri	Unable to verify the reconstructed waveform against the original signal.	20	2	

		and numerical error for each N. Shows that the error decreases as N increases, and identifies the Gibbs phenomenon at discontinuities with a correct explanation.	minor errors in calculation or interpretation of convergence.	cal error. Convergence or Gibbs phenomenon is not explained, and needs guidance.				
Interpret the Fourier series results and mainta	Engineering Analysis, Record &	Provides technically accurate interpr	Maintains a proper record with mostly accura	Record and interpretation are basic or	Incomplete record, limited or incorrect	20	2	

in the record.	Interpretation	etation of the results, relating the number of harmonics, symmetry and discontinuities to reconstruction accuracy. Maintains a complete, well-organized record with code, plots, observations and conclusions.	te interpretation, with minor omissions in observations or conclusions.	partly incomplete, with limited technical reasoning.	interpretation of the results.			
					Total	100	10	